

Formative Assessment 4

Dummy-Variable Regression Using Real-World Data

Dataset: diamonds **Total:** 40 points

Reference-category and coding choice

The reference category is **Ideal**. This lets every dummy coefficient be interpreted as a price difference relative to Ideal at the same carat. Because the assessment specifically requires dummy-variable regression, the R script explicitly converts *cut* to a regular factor with Ideal as its first level.

Part A. Exploring and Preparing the Data — 8 points

1. The dataset contains **53,940 observations** and **10 variables**.
2. The response variable is **price**, the price of a diamond in US dollars.
3. The quantitative explanatory variable is **carat**, the weight of the diamond.
4. The categorical explanatory variable is **cut**, with categories Fair, Good, Very Good, Premium, and Ideal.
5. **Ideal** is selected as the reference category. A reference category is needed so the other category coefficients are interpreted relative to one category and to avoid perfect multicollinearity.

Part B. Constructing Dummy Variables — 8 points

Cut	D_Fair	D_Good	D_VeryGood	D_Premium
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1
Ideal	0	0	0	0

2. The all-zero category is **Ideal**, the reference category.
3. Five dummy variables cannot be included together with an intercept because the five dummies sum to 1 for every observation. This creates exact linear dependence with the intercept (the dummy-variable trap), so the coefficients are not uniquely estimable.

Part C. Additive Dummy-Variable Regression — 10 points

Model: $\text{price} = \beta_0 + \beta_1 \text{carat} + \gamma_1 D_{\text{Fair}} + \gamma_2 D_{\text{Good}} + \gamma_3 D_{\text{VeryGood}} + \gamma_4 D_{\text{Premium}} + \epsilon_i$.

Term	Estimate	Std. Error	t value	p-value
Intercept	-2074.546	14.232	-145.769	<0.001
D_Fair	-1800.924	39.344	-45.773	<0.001
D_Good	-680.592	23.997	-28.362	<0.001
D_VeryGood	-290.789	17.239	-16.868	<0.001
D_Premium	-361.847	16.693	-21.677	<0.001

Carat	7871.082	13.980	563.040	<0.001
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1. Estimated equation:

$$\text{Price} = -2074.546 + 7871.082(\text{carat}) - 1800.924D_{\text{Fair}} - 680.592D_{\text{Good}} - 290.789D_{\text{VeryGood}} - 361.847D_{\text{Premium}}$$

2. Holding cut constant, a 1-carat increase is associated with an estimated **\$7,871.082** increase in diamond price.

3. At the same carat, Fair has an estimated price **\$1,800.924 lower** than Ideal. The other dummy coefficients are interpreted similarly relative to Ideal.

4. $R^2 = 0.856$, so approximately **85.6%** of the observed variation in price is explained by carat and cut in the additive model.

5. The additive model gives all cut categories the **same slope** with respect to carat; cut changes the intercepts only.

Part D. Interaction Between Carat and Cut — 10 points

1. H_0 : all four carat $\times$ cut interaction coefficients are 0. H_a : at least one interaction coefficient is not 0.

Interaction	Estimate	Std. Error	t value	p-value
carat $\times$ Fair	-2267.896	76.053	-29.820	8.195e-194
carat $\times$ Good	-712.755	52.670	-13.533	1.175e-41
carat $\times$ Very Good	-256.419	37.886	-6.768	1.317e-11
carat $\times$ Premium	-384.639	34.181	-11.253	2.414e-29

3. Nested-model ANOVA / incremental F-test: $F(4, 53930) = 247.735$, $p = 2.893e-211$.

4. At $\alpha = 0.05$, reject H_0 because $p = 2.893e-211 < 0.05$. The interaction terms contribute statistically significant additional information, indicating that the association between carat and price differs across cut categories.

5. The additive model assumes one common carat slope across cuts, while the interaction model permits a different caratslope for each cut category.

Part E. Visualization and Statistical Conclusion — 4 points



The scatterplot shows a strong positive relationship between carat and diamond price, with visible differences among cut categories. The cut dummy variables describe price differences relative to the Ideal reference category at a fixed carat. The additive model has $R^2 = 0.856$, meaning it explains about 85.6% of the observed price variation. The interaction test is statistically significant, showing that the effect of carat on price differs across cut categories. Thus, the common-slope assumption of the additive model does not fully describe the observed relationships. Overall, the regression analysis indicates that carat and cut, together with their interaction, are important statistical predictors of diamond price.

Reproducibility: R Commands Used

```
library(ggplot2)
data(diamonds)

head(diamonds)
str(diamonds)
summary(diamonds)

diamonds$cut <- factor( diamonds$cut, levels = c("Ideal",
"Fair", "Good", "Very Good", "Premium") )

levels(diamonds$cut)
contrasts(diamonds$cut)

dummy_table <- data.frame( Cut = c("Fair", "Good", "Very
Good", "Premium", "Ideal"),
  D_Fair = c(1, 0, 0, 0, 0),
  D_Good = c(0, 1, 0, 0, 0),
  D_VeryGood = c(0, 0, 1, 0, 0),
  D_Premium = c(0, 0, 0, 1, 0)
) dummy_table

modell <- lm(price ~ carat + cut, data = diamonds)
summary(modell)

model2 <- lm(price ~ carat * cut, data = diamonds)
summary(model2)

anova(modell, model2)

p <- ggplot(diamonds, aes(x = carat, y = price, color = cut)) +
geom_point(alpha = 0.3) + geom_smooth(method = "lm", se =
FALSE) + labs( title = "Diamond Price versus Carat by
Cut", x = "Carat", y = "Price (US Dollars)" )

p

ggsave( filename =
"figures/diamond_price_vs_carat_by_cut.png", plot = p,
width = 9, height = 6, dpi = 300 )
```

Selected numerical output

Quantity	Value
Observations	53,940
Variables	10
Additive R ²	0.856475
Interaction F	247.734792
Interaction df	4, 53930
Interaction p-value	2.892820e-211